

**DESIGN AND DEVELOPMENT OF AN AI-POWERED LEAD GENERATION
AND AUTONOMOUS OUTREACH PLATFORM USING
MULTI-AGENT SYSTEMS**

SUBMITTED BY

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25PCS205

Under the guidance of

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Submitted to

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FOR THE PARTIAL FULFILLMENT OF THE REQUIREMENTS FOR THE

AWARD OF THE DEGREE OF

M.Sc., COMPUTER SCIENCE



THE AMERICAN COLLEGE

(An Autonomous Institution Affiliated to Madurai Kamamj University)

Re-accredited with (3rd cycle) by NAAC with Grade “A+”, CGPA — 3.47 on a 4-Point scale MADURAI - 625002

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May - 2026



BONAFIDE CERTIFICATE

This is to certify that the project work entitled “DESIGN AND DEVELOPMENT OF AN AI-POWERED LEAD GENERATION AND AUTONOMOUS OUTREACH PLATFORM USING MULTI-AGENT SYSTEMS” is a bonafide record of work done by **Mogeshwaran M(25PCS205)** . In partial fulfillment of the requirements for the award of the degree of M.Sc., Computer Science of The American College . This work has not been submitted for the award of any degree to the best of our knowledge.

Submitted for viva-voice examination, held on _____ in the Department of Computer Science, The American College, Madurai.

Internal Guide

Head of the Department

INTERNSHIP ACCOMPLISHMENT CERTIFICATE

This is to certify that Mr. M. Mogeshwaran, a student of Master of Science (M.Sc.) in Computer Science at The American College, Madurai, has successfully completed his internship as an AI Product Developer Trainee in the AI Product Development & Management Division at PiBi Info Technologies LLP, Madurai, during the period from 26 November 2025 to 28 February 2026.

During the internship, he actively participated in the design, development, and implementation of AI-driven business solutions and modern web applications. He gained hands-on experience in Full Stack Development, Artificial Intelligence, Database Management, and Software Engineering practices.

As part of his internship responsibilities, he contributed to:

- Development of AI-powered Lead Automation Systems for lead qualification, assignment, and workflow automation.
- Design and implementation of Governance AI solutions for intelligent process management and decision support.
- Development and enhancement of CRM Engine modules for customer relationship management and business process optimization.
- Building and integration of AI Agents to automate operational and business workflows.
- Frontend development using Next.js and backend development using Python.
- Database design and management using Supabase and MySQL.
- API integration, application testing, debugging, and performance optimization.
- Technical documentation and participation in product development activities.

Throughout the internship period, Mr. Mogeshwaran demonstrated strong technical competence, analytical skills, dedication, and professionalism. He completed all assigned responsibilities successfully and made valuable contributions to the organization's AI product development initiatives.

We appreciate his commitment and sincere efforts during the internship and wish him continued success in his future academic and professional endeavors.

For PiBi Info Technologies LLP

Siranjeevi Ramdoss Chief
Executive Officer

PiBi Info Technologies
LLP Madurai,
Tamil Nadu
Date: 28 February 2026

DECLARATION

I hereby declare that the internship report entitled “**Development of AI-Powered Business Automation Solutions using AI Agents, Next.js, Python, Supabase, and MySQL**” is a bonafide record of the work carried out by me at **PiBi Info Technologies LLP, Madurai**, during the period from **26 November 2025 to 28 February 2026**, under the guidance and supervision of the concerned mentors of the organization.

I further declare that this report has been prepared by me in partial fulfillment of the requirements for the award of the degree of **Master of Science (M.Sc.) in Computer Science** at **The American College, Madurai**.

I also declare that this work is original and has not been submitted, either in part or in full, to any other University, Institution, or Organization for the award of any degree, diploma, certificate, or other qualification.

Place: Madurai

Date: _____

Signature of the Student

M. Mogeshwaran

M.Sc. Computer Science

The American College, Madurai

ACKNOWLEDGEMENT

I take this opportunity to express my heartfelt gratitude and sincere thanks to all those who have guided, supported, and encouraged me throughout the successful completion of my internship report entitled “**Development of AI-Powered Business Automation Solutions using AI Agents, Next.js, Python, Supabase, and MySQL.**”

I express my sincere gratitude to our **Principal & Secretary, Dr. J. Paul Jayakar, M.A., M.Phil., Ph.D., PBDCA.**, for providing me with the opportunity to undertake this internship and for creating an excellent academic environment that promotes learning and professional growth.

My special thanks are due to our **Head of the Department, Mr. K. Britto Alex, MCA., M.Phil., SET, (Ph.D.)**, for his constant encouragement, valuable suggestions, and support throughout the course of this internship work.

I also express my profound gratitude to my project guide **Mrs. Sankareswari K.**, for her continuous guidance, constructive feedback, and motivation, which greatly contributed to the successful completion of this internship report.

I would like to extend my sincere thanks to the management and technical team of **PiBi Info Technologies LLP, Madurai**, for providing me with an excellent opportunity to gain practical exposure in the field of Artificial Intelligence and Full Stack Development. I am especially grateful to **Mr. Siranjeevi Ramdoss, Chief Executive Officer**, and the mentors of the **AI Product Development & Management Division** for their guidance and support during the internship period.

During the internship, I had the opportunity to work on advanced technologies including **Next.js, Python, Supabase, MySQL, AI Agents, Lead Automation Systems, Governance AI Solutions, and CRM Engine Development**, which significantly enhanced my technical knowledge and professional skills.

Finally, I express my sincere thanks to all the faculty members, friends, family members, and everyone who directly or indirectly contributed to the successful completion of this internship report.

M. Mogeshwaran

M.Sc. Computer Science

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ABSTRACT

This internship report presents the work carried out during my internship as an **AI Product Developer Trainee** at **PiBi Info Technologies LLP, Madurai**, from **26 November 2025 to 28 February 2026**. The internship provided practical exposure to Artificial Intelligence, Full Stack Development, Database Management, and Business Process Automation.

The primary objective of the internship was to understand and implement AI-driven solutions that enhance business productivity and automate operational workflows. During the internship, I worked on the development of **AI-powered Lead Automation Systems, Governance AI Solutions, CRM Engine Development, and AI Agent Integration**. These solutions were designed to streamline lead management, automate business processes, improve customer relationship management, and provide intelligent decision-support capabilities.

The technical implementation involved the use of modern technologies such as **Next.js** for frontend development, **Python** for backend services and AI workflows, **Supabase** and **MySQL** for database management, and AI Agents for automation and intelligent task execution. The internship also provided hands-on experience in API integration, system design, testing, debugging, deployment support, and technical documentation.

Through this internship, I gained valuable knowledge of software development life cycle practices, AI product development methodologies, cloud-based database systems, and enterprise-level application development. The experience significantly enhanced my technical, analytical, and problem-solving skills while providing practical exposure to real-world industry requirements.

The successful completion of this internship has strengthened my understanding of Artificial Intelligence and Full Stack Development and has prepared me for future opportunities in Software Engineering, AI Development, and Product Engineering.

Keywords: Artificial Intelligence, AI Agents, Lead Automation, Governance AI, CRM Engine, Next.js, Python, Supabase, MySQL, Full Stack Development, Business Automation.

1.INTRODUCTION

1.1 Overview of Artificial Intelligence

Artificial Intelligence (AI) is one of the most rapidly growing fields in computer science, enabling machines and software systems to perform tasks that traditionally require human intelligence. AI technologies are capable of learning from data, recognizing patterns, making decisions, automating repetitive tasks, and providing intelligent solutions to complex business problems. In recent years, AI has become an integral part of various industries including healthcare, finance, education, manufacturing, e-commerce, and customer relationship management.

The increasing adoption of AI technologies has transformed the way organizations operate by improving efficiency, reducing operational costs, and enhancing decision-making capabilities. Modern AI systems utilize machine learning, natural language processing, automation, and intelligent agents to streamline business operations and deliver innovative solutions.

1.2 Importance of Business Automation

Business automation refers to the use of technology to automate repetitive and time-consuming business processes with minimal human intervention. Organizations are increasingly adopting automation solutions to improve productivity, reduce errors, and optimize resource utilization.

AI-powered automation systems can perform tasks such as lead management, customer support, workflow monitoring, data processing, and decision support. These systems help organizations respond quickly to customer needs, improve operational efficiency, and achieve better business outcomes.

Automation also enables businesses to scale their operations while maintaining consistency and accuracy. As a result, business automation has become a critical component of modern enterprise software solutions.

1.3 Need for AI Agents

AI Agents are intelligent software entities capable of performing specific tasks autonomously based on predefined goals and contextual information. These agents can analyze data, interact with users, make recommendations, execute workflows, and automate business processes.

The demand for AI Agents has significantly increased due to their ability to reduce manual effort and improve operational efficiency. Organizations use AI Agents for customer support, lead qualification, workflow automation, governance monitoring, data analysis, and decision-making processes.

During the internship, AI Agents were utilized in various business applications including Lead Automation Systems, Governance AI Solutions, and CRM Engine development. These agents contributed to enhanced productivity, faster response times, and improved.

2. ORGANIZATION PROFILE

2.1 Company Overview

PiBi Info Technologies LLP is a technology-driven organization headquartered in Madurai, Tamil Nadu, specializing in software development, Artificial Intelligence solutions, enterprise applications, business automation systems, and digital transformation services. The company focuses on delivering innovative technology solutions that help organizations improve operational efficiency, productivity, and decision-making processes.

PiBi Info Technologies provides industry-oriented training and practical exposure to students and professionals through real-time projects and product development activities. The organization promotes innovation and encourages the adoption of emerging technologies such as Artificial Intelligence, Machine Learning, Cloud Computing, and Full Stack Development.

2.2 Services Offered

PiBi Info Technologies provides a wide range of technology services, including:

- **Software Development**
Development of custom software applications tailored to business requirements.
- **Web Application Development**
Design and development of responsive and scalable web applications using modern technologies.
- **Artificial Intelligence Solutions**
Implementation of AI-powered systems, intelligent automation, predictive analytics, and AI Agents.
- **Business Process Automation**
Development of automation solutions to streamline workflows and improve operational efficiency.
- **Customer Relationship Management Solutions**
Development and enhancement of CRM platforms for customer engagement and lead management.
- **Database Management Solutions**
Design, implementation, and maintenance of secure and scalable database systems.
- **Technology Training and Internship Programs**
Providing industry-oriented training programs and internship opportunities for students and graduates.

2.3 AI Product Development & Management Division

The AI Product Development & Management Division focuses on designing, developing, and managing Artificial Intelligence-based products and enterprise solutions. The division is responsible for:

- AI Product Design and Development
- AI Agent Development
- Business Process Automation
- Workflow Optimization
- CRM Solution Development
- Data Analytics and Reporting
- AI Model Integration
- Product Testing and Deployment

During the internship, I worked under this division as an **AI Product Developer Trainee**, gaining practical exposure to AI-powered product development and modern software engineering practices.

2.4 Internship Environment

The internship environment at PiBi Info Technologies was highly professional and learning-oriented. The organization encouraged innovation, teamwork, and continuous learning through real-time projects and technical discussions.

The internship provided opportunities to work with advanced technologies such as:

- Next.js
- Python
- Supabase
- MySQL
- REST APIs
- AI Agents
- Lead Automation Systems
- Governance AI Solutions
- CRM Engine Development

The experience helped bridge the gap between academic learning and industry requirements by providing hands-on exposure to enterprise software development and Artificial Intelligence applications.

3. SYSTEM STUDY

3.1 Existing System

Many organizations manage leads, customer interactions, approvals, and operational workflows manually or through disconnected software systems. Manual processing of leads and customer information often results in delays, data duplication, reduced productivity, and inefficient decision-making.

Traditional Customer Relationship Management (CRM) systems provide basic customer management capabilities but often lack intelligent automation and AI-driven insights.

Similarly, governance and workflow management processes require significant human intervention, making them time-consuming and prone to errors.

The absence of intelligent automation leads to:

- Delayed lead response and conversion.
- Inefficient customer management.
- Manual approval and governance processes.
- Increased operational workload.
- Limited decision-support capabilities.
- Poor workflow visibility and tracking.

3.2 Proposed System

The proposed system introduces Artificial Intelligence-powered automation solutions to improve business operations and customer management processes.

The solution integrates AI Agents, Lead Automation, Governance AI, and CRM functionalities into a unified platform capable of automating repetitive tasks and providing intelligent assistance.

The proposed system is developed using modern technologies such as Next.js, Python, Supabase, and MySQL.

The system provides:

- Automated lead capture and qualification.
- AI-powered customer relationship management.
- Intelligent workflow and governance automation.
- AI Agent-based task execution.
- Real-time data management and reporting.
- Enhanced business productivity and efficiency.

3.3 Objectives of the System

The major objectives of the proposed system are:

- To automate lead management processes.
- To improve customer relationship management.
- To reduce manual effort through AI Agents.
- To streamline governance and approval workflows.

3.4 Advantages of the Proposed System

The proposed system offers several advantages:

- Faster lead processing.
- Improved customer engagement.
- Reduced operational costs.
- Increased productivity.
- Better workflow management.
- Accurate data handling.
- Scalable architecture.
- AI-powered automation capabilities.

3.5 Software Requirements

Frontend

- Next.js
- HTML
- CSS
- JavaScript

Backend

- Python

Database

- Supabase
- MySQL

Tools

- Visual Studio Code
- Git
- Web Browser

3.6 Hardware Requirements

Minimum Requirements

- Processor : Intel Core i3 / Ryzen 3
- RAM : 4 GB
- Storage : 128 GB SSD
- Internet Connection

Recommended Requirements

- Processor : Intel Core i5 / Ryzen 5 or above
- RAM : 8 GB or above
- Storage : 256 GB SSD or above
- Stable Internet Connection

4. SYSTEM DESIGN

4.1 System Architecture

The AI-Powered Business Automation System is designed using a modern full-stack architecture that integrates frontend, backend, database, and Artificial Intelligence components.

The system consists of the following layers:

- **Presentation Layer**

The Presentation Layer is developed using **Next.js**, which provides a responsive and user-friendly interface for users to interact with the system.

- **Application Layer**

The Application Layer is implemented using **Python**, which handles business logic, workflow processing, API communication, and AI Agent execution.

- **Database Layer**

The Database Layer utilizes **Supabase** and **MySQL** for secure storage, retrieval, and management of application data.

- **Artificial Intelligence Layer**

The AI Layer consists of intelligent AI Agents responsible for lead qualification, workflow automation, governance monitoring, and CRM assistance.

The architecture ensures scalability, security, and efficient communication between different modules of the system.

4.2 Module Description

The system is divided into several functional modules.

4.2.1 Lead Automation Module

This module automates lead management processes from lead capture to conversion.

Functions:

- Lead collection
- Lead qualification
- Lead assignment
- Lead tracking
- Automated follow-up
- Lead analytics

4.2.2 Governance AI Module

This module automates governance-related workflows and organizational processes.

Functions:

- Policy monitoring
- Workflow approvals
- Process automation

4.2.3 CRM Engine Module

This module manages customer information and interactions.

Functions:

- Customer profile management
- Lead management
- Customer activity tracking
- Interaction history
- Business reporting

4.2.4 AI Agent Module

This module contains intelligent AI Agents capable of executing tasks automatically.

Functions:

- Automated task execution
- Query processing
- Workflow assistance
- Intelligent recommendations
- Business process automation

4.2.5 User Management Module

This module handles authentication and authorization.

Functions:

- User registration
- Login and authentication
- Role management
- Access control
- Security management

4.3 Data Flow

The data flow of the system follows the sequence below:

1. User submits information through the web interface.
2. Next.js frontend sends requests to Python APIs.
3. Backend processes the request.
4. AI Agents analyze and perform required operations.
5. Data is stored and retrieved from Supabase and MySQL.
6. Processed results are returned to the frontend.
7. Users receive real-time updates and reports.

4.4 Database Design

The database is designed to store application information efficiently.

Main Tables

1. User Table

Stores:

- User ID
- Name
- Email
- Role
- Password

2. Lead Table

Stores:

- Lead ID
- Lead Name
- Contact Details
- Status
- Assigned User
- Follow-up Information

3. Customer Table

Stores:

- Customer ID
- Customer Information
- Interaction History
- Activity Records

4. Governance Table

Stores:

- Process Information
- Approval Status
- Compliance Records
- Workflow Data

5. AI Agent Activity Table

Stores:

- Agent ID
- Task Details
- Execution Status
- Timestamp
- Output Results

4.5 Security Features

The system incorporates multiple security mechanisms:

- User Authentication
- Role-Based Access Control
- Secure API CommunicationDatabase Protection

5. IMPLEMENTATION

5.1 Introduction

Implementation is the process of transforming system requirements and designs into a working software solution. During the internship at PiBi Info Technologies LLP, various AI-powered business automation solutions were developed using modern technologies such as Next.js, Python, Supabase, MySQL, and AI Agents. The implementation phase focused on developing scalable, efficient, and intelligent systems capable of automating business processes and improving organizational productivity.

5.2 Lead Automation System Implementation

The Lead Automation System was developed to automate the lead management process from lead generation to lead conversion.

Features Implemented

- Lead Registration
- Lead Qualification
- Lead Assignment
- Lead Status Tracking
- Follow-up Management
- Lead Analytics

Working Process

1. Leads are collected through various sources.
2. Lead information is stored in the database.
3. AI Agents analyze lead data.
4. Qualified leads are assigned automatically.
5. Follow-up reminders are generated.
6. Lead status is continuously monitored.

Benefits

- Reduced manual effort.
- Faster lead response.
- Improved lead conversion rate.
- Better tracking and reporting.

5.3 Governance AI Implementation

Governance AI was developed to automate internal organizational workflows and improve decision-making processes.

Features Implemented

- Workflow Monitoring
- Automated Approvals
- Compliance Tracking
- Task Management
- Process Optimization

Working Process

1. Business processes are configured.
2. AI Agents monitor workflow activities.
3. Rules and conditions are evaluated.
4. Automated actions are triggered.
5. Reports are generated for management review.

Benefits

- Increased operational efficiency.
- Improved transparency.
- Reduced processing time.
- Better compliance management.

5.4 CRM Engine Implementation

The CRM Engine was developed to manage customer relationships and business interactions effectively.

Features Implemented

- Customer Profile Management
- Lead Management
- Activity Tracking
- Communication History
- Business Reporting

Working Process

1. Customer data is collected.
2. Information is stored securely.
3. User interactions are recorded.
4. Activities are tracked automatically.
5. Reports and insights are generated.

Benefits

- Improved customer engagement.
- Better relationship management.
- Centralized customer information.
- Enhanced business productivity.

5.5 AI Agent Development

AI Agents were developed to automate repetitive tasks and assist users in performing business operations efficiently.

Features Implemented

- Intelligent Task Execution
- Workflow Automation
- Information Retrieval
- Recommendation Generation
- Process Monitoring

Working Process

1. User requests are received.
2. AI Agents analyze the request.
3. Required actions are executed.
4. Results are generated and returned.
5. Activities are logged for future reference.

Benefits

- Reduced manual intervention.
- Faster task completion.
- Improved operational efficiency.
- Intelligent automation support.

5.6 Frontend Development using Next.js

The frontend application was developed using Next.js to provide a responsive and user-friendly interface.

Features

- Dynamic User Interface
- Component-Based Architecture
- API Integration
- Responsive Design
- Secure User Authentication

Advantages

- High Performance
- Faster Loading Speed
- Better User Experience
- Scalability

5.7 Backend Development using Python

Python was used for implementing business logic, APIs, workflow processing, and AI-related functionalities.

Features

- REST API Development
- Data Processing
- AI Integration
- Business Logic Implementation
- Database Connectivity

Advantages

- Easy Maintenance
- High Flexibility
- Efficient Processing
- Extensive Library Support

5.8 Database Implementation

The system uses Supabase and MySQL for data storage and management.

Supabase

Used for:

- Authentication
- Real-Time Database Operations
- Cloud Data Storage

MySQL

Used for:

- Structured Data Storage
- Query Processing
- Data Management

Advantages

- Secure Data Storage
- High Availability
- Reliable Performance
- Scalability

5.9 Summary

The implementation phase successfully delivered AI-powered business automation solutions capable of improving lead management, governance processes, customer relationship management, and workflow automation. The integration of AI Agents, modern web technologies, and cloud databases resulted in a scalable and efficient system suitable for real-world business environments.

6. TESTING AND RESULTS

6.1 Introduction

Testing is an essential phase in software development that ensures the developed system functions correctly and satisfies user requirements. The AI-Powered Business Automation System was thoroughly tested to verify the functionality, performance, reliability, and accuracy of all modules.

The testing process focused on validating the Lead Automation System, Governance AI Module, CRM Engine, AI Agents, frontend interfaces, backend APIs, and database operations.

6.2 Testing Objectives

The objectives of testing were:

- To verify that all modules function correctly.
- To identify and eliminate software defects.
- To ensure proper communication between system components.
- To validate database operations.
- To test AI Agent functionality.
- To improve system reliability and performance.

6.3 Types of Testing Performed

1. Unit Testing

Individual components were tested independently to verify their functionality.

Modules Tested:

- Lead Management Module
- Governance AI Module
- CRM Engine
- AI Agent Module
- User Authentication Module

2. Integration Testing

Integration testing was performed to ensure proper interaction between:

- Next.js Frontend
- Python Backend
- Supabase Database
- MySQL Database
- AI Agents

3. System Testing

The complete application was tested as an integrated system to verify end-to-end functionality.

4. User Acceptance Testing

The developed features were tested from the user's perspective to ensure that the system met business requirements.

6.4 Test Cases

Test Case 1

Module: User Authentication

Test Scenario	Expected Result	Status
Valid Login Credentials	User Login Successful	Passed
Invalid Credentials	Error Message Displayed	Passed

Test Case 2

Module: Lead Automation

Test Scenario	Expected Result	Status
New Lead Creation	Lead Stored Successfully	Passed
Lead Assignment	Lead Assigned Correctly	Passed
Lead Status Update	Status Updated Successfully	Passed

Test Case 3

Module: CRM Engine

Test Scenario	Expected Result	Status
Customer Creation	Customer Added Successfully	Passed
Customer Update	Customer Information Updated	Passed
Activity Tracking	Activities Recorded Properly	Passed

Test Case 4

Module: Governance AI

Test Scenario	Expected Result	Status
Workflow Submission	Process Initiated Successfully	Passed
Approval Automation	Approval Generated Correctly	Passed
Status Tracking	Workflow Status Updated	Passed

Test Case 5

Module: AI Agents

Test Scenario	Expected Result	Status
Task Execution	Task Completed Successfully	Passed
Workflow Trigger	Action Executed Correctly	Passed
Response Generation	Correct Response Returned	Passed

6.5 Results Obtained

The developed system successfully achieved its intended objectives.

Lead Automation Results

- Faster lead processing.
- Reduced manual lead handling.
- Improved lead tracking.
- Better lead management efficiency.

Governance AI Results

- Automated workflow management.
- Reduced approval delays.
- Improved process monitoring.
- Enhanced operational transparency.

CRM Engine Results

- Efficient customer data management.
- Improved customer interaction tracking.
- Centralized business information.
- Better reporting capabilities.

AI Agent Results

- Automated repetitive business tasks.
- Reduced manual workload.
- Improved operational productivity.
- Faster information retrieval and processing.

6.6 Performance Analysis

The system demonstrated satisfactory performance in terms of:

- Response Time
- Data Processing Speed
- Database Operations
- Workflow Execution
- System Reliability

The use of Next.js, Python, Supabase, and MySQL contributed to efficient system performance and scalability.

6.7 Screenshots

The screenshots of the developed modules and interfaces are provided in the Appendix section.

Screenshots include:

- Dashboard
- Lead Automation Module
- CRM Engine
- AI Agent Interface

7.CONCLUSION AND FUTURE ENHANCEMENT

7.1 Conclusion

The internship at **PiBi Info Technologies LLP, Madurai** provided an excellent opportunity to gain practical exposure to modern software development technologies and Artificial Intelligence applications. The internship enabled me to bridge the gap between academic learning and real-world industry practices.

During the internship period, I actively participated in the development of AI-powered business automation solutions, including **Lead Automation Systems, Governance AI Solutions, CRM Engine Development, and AI Agent Integration**. These projects helped me understand the complete software development lifecycle, from requirement analysis and system design to implementation, testing, and deployment. The internship also provided hands-on experience with modern technologies such as **Next.js, Python, Supabase, and MySQL**, which significantly improved my technical knowledge and practical development skills. Working on real-world business problems enhanced my problem-solving abilities, analytical thinking, teamwork, and professional communication skills.

The successful implementation of AI-powered solutions demonstrated how Artificial Intelligence can improve operational efficiency, automate repetitive tasks, streamline workflows, and support business decision-making processes. Overall, the internship was highly beneficial and contributed significantly to my academic growth, technical expertise, and professional development.

7.2 Learning Outcomes

Through this internship, I gained practical knowledge and experience in:

- Artificial Intelligence and AI Agents
- Full Stack Web Development
- Next.js Application Development
- Python Backend Development
- API Development and Integration
- Database Design and Management
- Supabase and MySQL
- CRM Systems
- Business Process Automation
- Governance AI Solutions
- Software Testing and Debugging
- Technical Documentation
- Team Collaboration and Professional Practices

The internship helped me strengthen both technical and professional skills required for a successful career in Software Engineering and Artificial Intelligence.

7.3 Future Enhancement

The developed system can be further enhanced by implementing advanced features and technologies.

Future enhancements may include:

Advanced AI Capabilities

- Integration of Large Language Models (LLMs)
- Intelligent Business Assistants
- Advanced Decision Support Systems
- Predictive Analytics

Enhanced Automation

- Multi-Agent Collaboration
- Automated Workflow Generation
- Smart Notification Systems
- Intelligent Task Prioritization

CRM Improvements

- Customer Behavior Analysis
- Predictive Lead Scoring
- Customer Sentiment Analysis
- Personalized Recommendations

Governance AI Enhancements

- Automated Compliance Auditing
- Risk Detection Systems
- Intelligent Policy Monitoring
- AI-Based Reporting

Technical Improvements

- Cloud Deployment
- Microservices Architecture
- Advanced Security Features
- Mobile Application Integration
- Real-Time Analytics Dashboard

These enhancements can further improve system efficiency, scalability, and business value while enabling organizations to leverage the full potential of Artificial Intelligence technologies.

7.4 Final Remark

The internship experience at PiBi Info Technologies LLP has provided valuable industry exposure and practical knowledge in Artificial Intelligence, Full Stack Development, and Business Automation. The skills acquired during this period will serve as a strong foundation for future academic pursuits and professional growth in the field of Computer Science and Artificial Intelligence.

PROJECTS

1. AI LEAD OUTREACH AGENT

GitHub link : https://github.com/mogesh-developer/ai_lead.git

1.1 Project Overview

The AI Lead Outreach Agent is an AI-powered business automation platform developed to automate the complete lead generation and outreach process. The system is capable of discovering potential business leads, analyzing their credibility, generating personalized outreach messages, and managing follow-up activities through autonomous AI Agents. The primary objective of the system is to reduce manual effort in lead generation and customer acquisition while improving outreach efficiency through Artificial Intelligence. The platform combines web scraping, AI analysis, lead scoring, autonomous decision-making, and automated communication into a single integrated solution.

1.2 Technologies Used

Frontend

- React.js
- Tailwind CSS
- Axios
- Vite

Backend

- Python
- Flask

Database

- MySQL

Artificial Intelligence

- Google Gemini 1.5 Flash
- Groq LLaMA-3

Web Scraping

- Playwright
- BeautifulSoup

Search Engine Integration

- DuckDuckGo Search API
-

1.3 System Modules

Lead Ingestion Module

This module allows users to upload lead information through CSV and Excel files.

Functions:

- Bulk Lead Import
 - Lead Validation
 - Data Storage
 - Lead Categorization
-

Lead Discovery Module

This module automatically searches the internet for potential business leads based on industry and location.

Functions:

- AI-Based Search
 - Business Discovery
 - Contact Information Collection
 - Lead Identification
-

Web Scraping Module

The Web Scraping Module extracts business information from websites.

Functions:

- Website Crawling
 - Email Extraction
 - Phone Number Extraction
 - Business Information Collection
-

AI Analysis Module

The AI Analysis Module evaluates lead quality using Artificial Intelligence.

Functions:

- Trust Score Generation
 - Business Analysis
 - Lead Qualification
 - Risk Assessment
-

Outreach Automation Module

This module generates personalized outreach messages automatically.

Functions:

- Message Generation
 - Email Personalization
 - Automated Communication
 - Follow-up Scheduling
-

1.4 Multi-Agent Architecture

The system consists of multiple specialized AI Agents working together.

- **Lead Discovery Agent**
Responsible for finding potential leads from online sources.
 - **Web Scraping Agent**
Collects business and contact information from websites.
 - **Business Analysis Agent**
Uses AI models to evaluate lead quality and trustworthiness.
 - **Decision Agent**
Determines whether a lead should be contacted.
 - **Message Strategy Agent**
Creates outreach strategies based on lead characteristics.
 - **Message Generation Agent**
Generates personalized outreach messages.
 - **Outreach Agent**
Sends emails automatically through SMTP services.
 - **Follow-up Agent**
Tracks communication and schedules future interactions.
-

1.5 System Workflow

1. User uploads leads or initiates AI lead discovery.
 2. Lead Discovery Agent searches for business leads.
 3. Web Scraping Agent extracts contact information.
 4. Business Analysis Agent evaluates lead quality.
 5. Decision Agent prioritizes leads.
 6. Message Generation Agent creates personalized messages.
 7. Outreach Agent sends emails automatically.
 8. Follow-up Agent manages future communication.
 9. Results are displayed on the dashboard.
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1.6 Key Features

- AI-Powered Lead Discovery
 - Intelligent Lead Scoring
 - Automated Outreach
 - Thanglish Message Generation
 - Real-Time Dashboard
 - AI Agent Collaboration
 - Automated Follow-Up System
 - Activity Monitoring
 - Performance Analytics
-

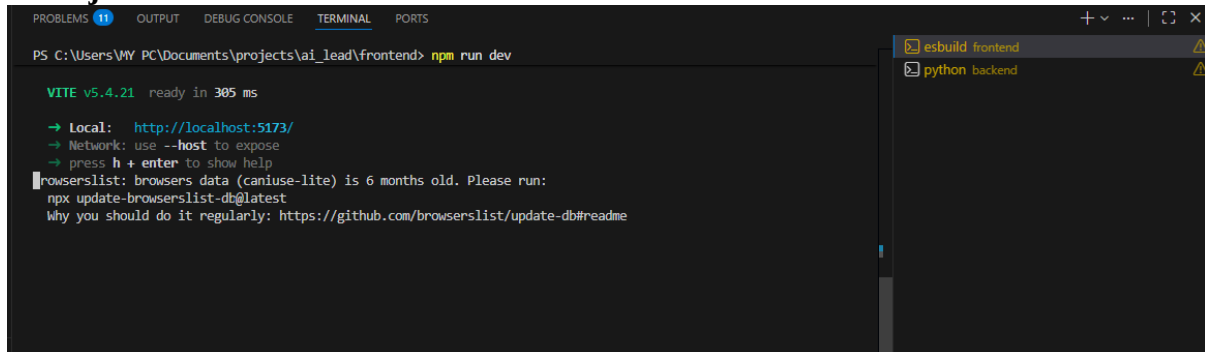
1.7 Advantages

- Reduces manual lead generation effort.
 - Improves outreach efficiency.
 - Enhances lead conversion opportunities.
 - Automates repetitive business tasks.
 - Provides intelligent business insights.
 - Supports scalable customer acquisition processes.
-

SCREENSHOT

Terminal:

Reactjs:



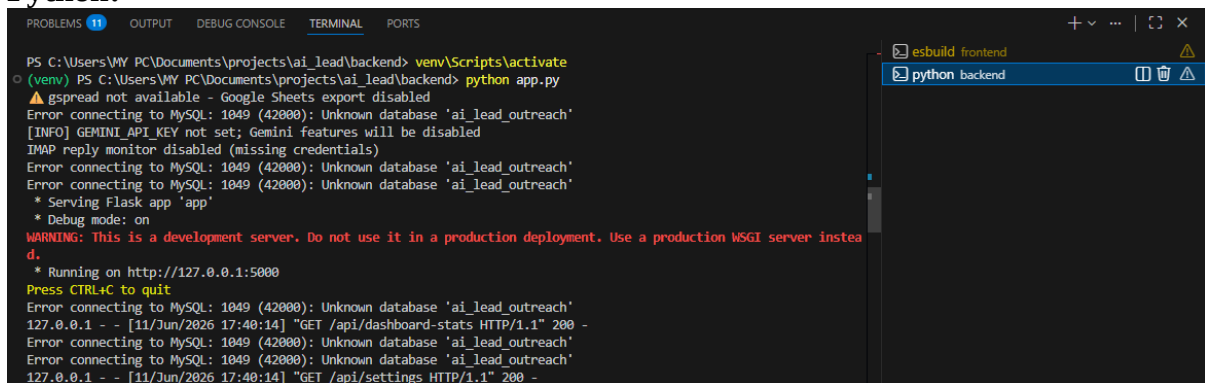
```
PS C:\Users\MY PC\Documents\projects\ai_lead\frontend> npm run dev

VITE v5.4.21 ready in 305 ms

  → Local:   http://localhost:5173/
  → Network: use --host to expose
  → press h + enter to show help

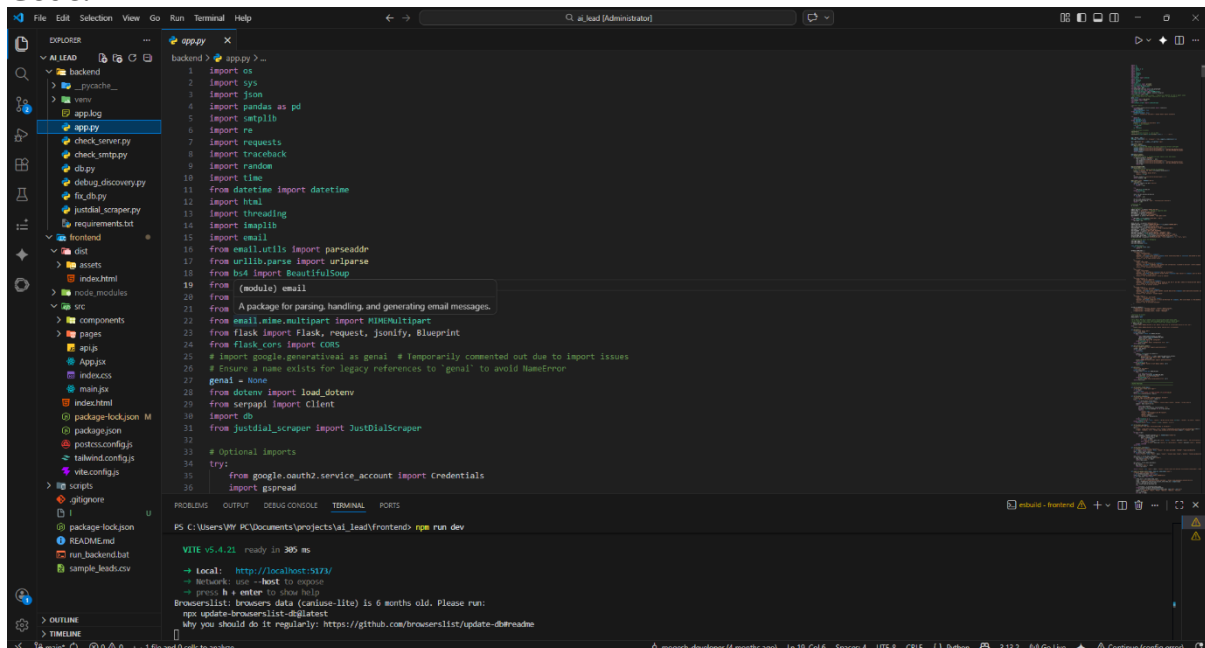
Browserslist: browsers data (caniuse-lite) is 6 months old. Please run:
  npx update-browserslist-db@latest
  Why you should do it regularly: https://github.com/browserslist/update-db#readme
```

Python:



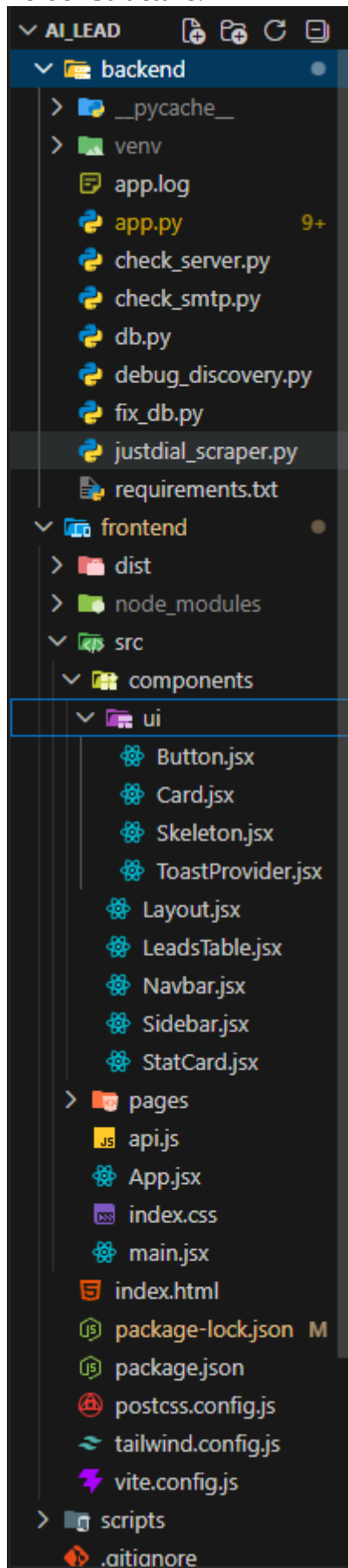
```
PS C:\Users\MY PC\Documents\projects\ai_lead\backend> venv\Scripts\activate
(venv) PS C:\Users\MY PC\Documents\projects\ai_lead\backend> python app.py
▲ gspread not available - Google Sheets export disabled
Error connecting to MySQL: 1049 (42000): Unknown database 'ai_lead_outreach'
[INFO] GEMINI_API_KEY not set; Gemini features will be disabled
IMAP reply monitor disabled (missing credentials)
Error connecting to MySQL: 1049 (42000): Unknown database 'ai_lead_outreach'
Error connecting to MySQL: 1049 (42000): Unknown database 'ai_lead_outreach'
* Serving Flask app 'app'
* Debug mode: on
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
* Running on http://127.0.0.1:5000
Press CTRL+C to quit
Error connecting to MySQL: 1049 (42000): Unknown database 'ai_lead_outreach'
127.0.0.1 - - [11/Jun/2026 17:40:14] "GET /api/dashboard-stats HTTP/1.1" 200 -
Error connecting to MySQL: 1049 (42000): Unknown database 'ai_lead_outreach'
Error connecting to MySQL: 1049 (42000): Unknown database 'ai_lead_outreach'
127.0.0.1 - - [11/Jun/2026 17:40:14] "GET /api/settings HTTP/1.1" 200 -
```

Code:

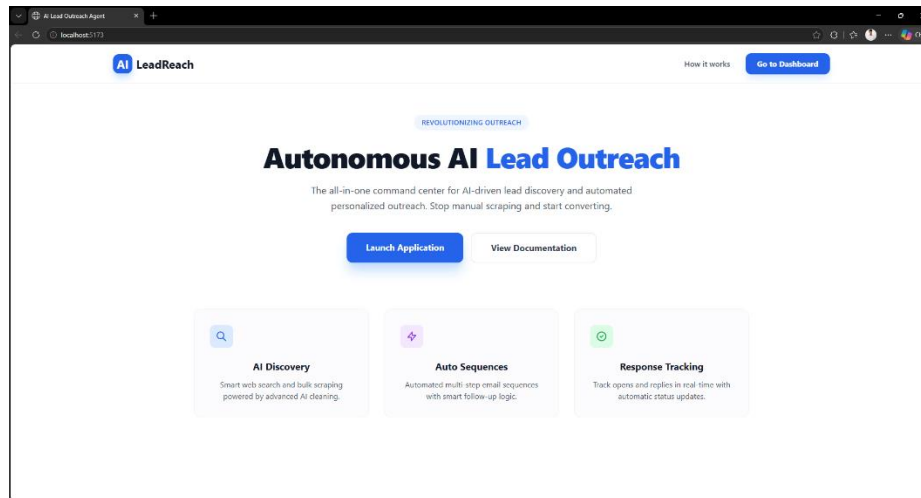


```
backend > app.py
1 import os
2 import sys
3 import json
4 import pandas as pd
5 import catpill
6 import re
7 import requests
8 import traceback
9 import random
10 import time
11 from datetime import datetime
12 import html
13 import threading
14 import imaplib
15 import email
16 from email.utils import parseaddr
17 from urllib.parse import urlparse
18 from bs4 import BeautifulSoup
19 from (module) email
20 from (module) email
21 from (module) email
22 from email.mime.multipart import MIMEMultipart
23 from flask import Flask, request, jsonify, Blueprint
24 from flask_cors import CORS
25 # import google.generativeai as genai # Temporarily commented out due to import issues
26 # Ensure a name exists for legacy references to 'genai' to avoid NameError
27 genai = None
28 from dotenv import load_dotenv
29 from serpapi import Client
30 import db
31 from JustDialScraper import JustDialScraper
32
33 # Optional imports
34 try:
35     from google.oauth2.service_account import Credentials
36     import gspread
```

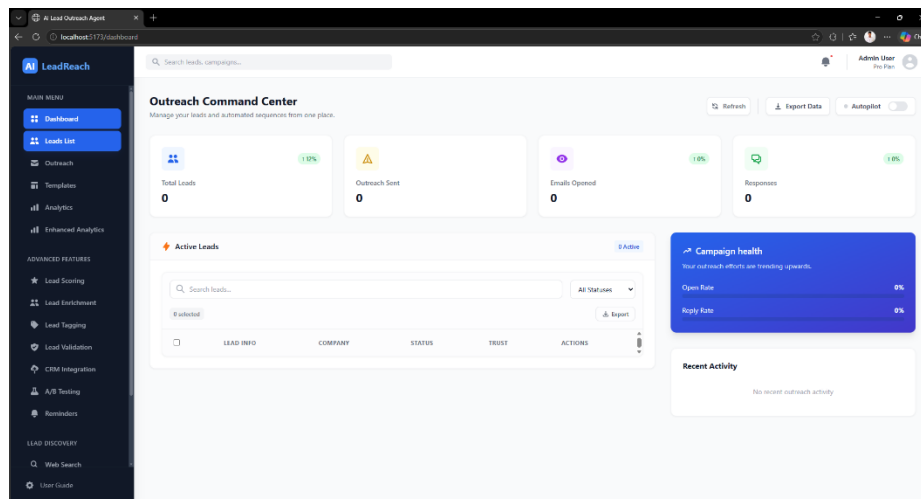
Folder Structure:



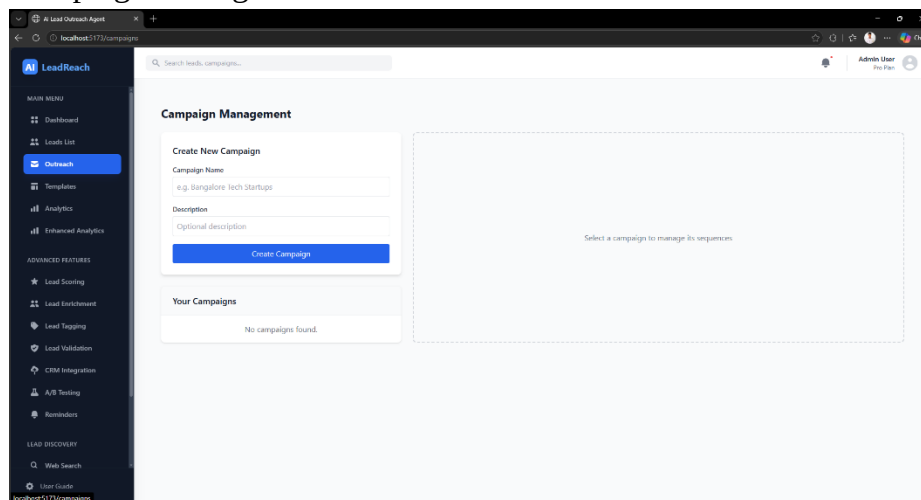
Home Page:



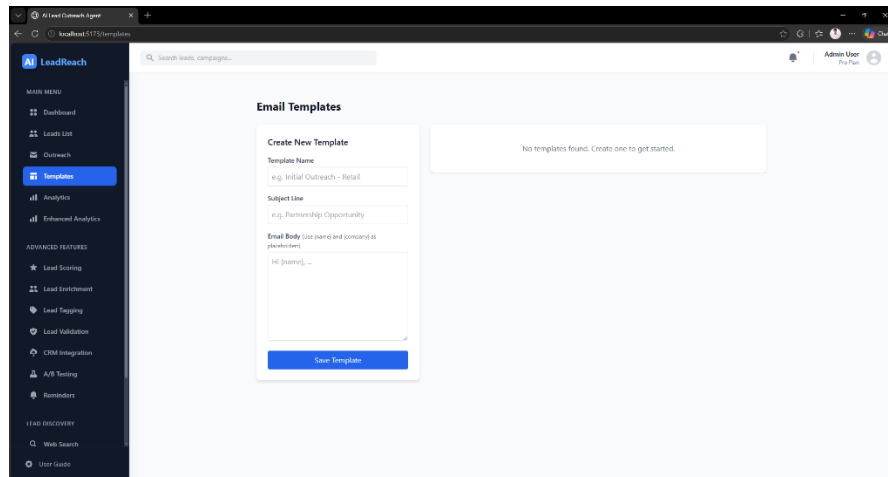
Dashboard:



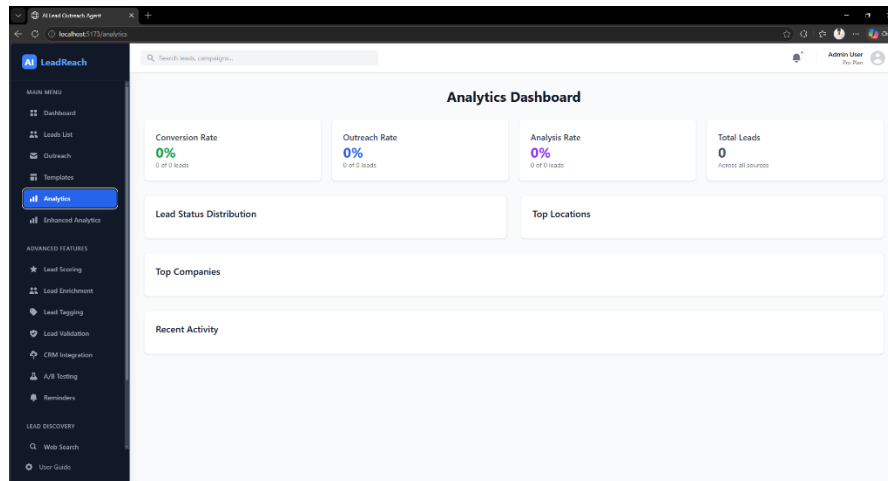
Campaign Management:



Outreach Email Management:



Analytics Dashboard:



Lead Searching:

